# Review for the nonlinear acoustics final

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*These problems, based on Dr. Hamilton's lectures, address the major topics of the latter half of the course, corresponding to HW6-HW8. Good luck on the exam!*

## 1 Rankine-Hugoniot relations

(a) Name the quantity that is conserved when $f$ and $g$, as defined below, are substituted into equation (1.1).

$$\frac{\partial f}{\partial t} + \frac{\partial g}{\partial x} = 0\,. \tag{1.1}$$

| | $f$ | $g$ |
|---|---|---|
| *(i)* | $\rho$ | $\rho u$ |
| *(ii)* | $\rho u$ | $\rho u^2 + P$ |
| *(iii)* | $\frac{1}{2}\rho u^2 + \rho e$ | $\frac{1}{2}\rho u^3 + \rho u e + P u$ |

(b) Write equation (1.1) in integral form by integrating from $x_1$ to $x_2$. Write the result such that the quantity $g(x_1, t) - g(x_2, t)$ appears on one side of the equation. Call this quantity $I$.

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(c) Let a discontinuity exist at $x_{\rm sh}(t)$ in the result from part (b). Split the integral $I$ from part (b) into $I_1 + I_2$, to account for the discontinuity. Use the notation $x_1 < x_{\rm sh}^- < x_{\rm sh}^+ < x_2$. *Hint: the upper limit of $I_1$ should be $x_{sh}^-$, and the lower limit of $I_2$ should be $x_{sh}^+$, where $x_{sh}^\pm = x_{sh} \pm \epsilon$, $\epsilon \to 0$.*

(d) Evaluate the integrals $I_1$ and $I_2$. *Hint: Note that for an arbitrary function $q(x,t)$,*

$$\frac{d}{dt}\int_{x_i}^{x^\pm} q(x,t)dx = q(x^\pm,t)\frac{dx^\pm}{dt} - q(x_i,t)\frac{dx_i}{dt} + \int_{x_i}^{x^\pm}\frac{\partial q}{\partial t}dx.$$

*Note that $dx_1(t)/dt = dx_2(t)/dt = 0$. Also, denote $dx_{sh}^-/dt = U_{sh}$.*

(e) Take the limit of $I_1$, as found in the previous part, as $x_1 \to x_{\rm sh}^-$. Similarly, take the limit of $I_2$ as $x_2 \to x_{\rm sh}^+$. Note that the integral vanishes in both cases.

(f) Use the above result, as well as the result of part (b), to show that as $x_1 \to x_{\rm sh}^-$ and $x_2 \to x_{\rm sh}^+$,

$$g(x_{\rm sh}^-,t) - g(x_{\rm sh}^+,t) = U_{\rm sh}[f(x_{\rm sh}^-,t) - f(x_{\rm sh}^+,t)] \tag{1.2}$$

(g) Rewrite equation (1.2) by letting the subscript $a$ correspond to "ahead of the shock," $x_{\rm sh}^+$, and by letting the subscript $b$ correspond to "behind the shock," $x_{\rm sh}^-$.

(h) Write the above result using the jump notation, $[q] = q_b - q_a$.

(i) Define $v = u - U_{\rm sh}$ and use the table from part (a) to derive the so-called Rankine-Hugoniot relations. *Hint: for the conservation of momentum and energy, some rearrangement is required. Just see your notes.*

(j) What did we find in class to be the order of the entropy jump across the shock, for an arbitrary fluid?

(k) What did we find in class to be the order of the reflection from the shock front?

(l) What do the previous two parts imply about the shock at quadratic order?

## 2 Weak shock speed

(a) Denoting $v = u - U_{\rm sh}$ and $Q = \rho u$, use the Rankine-Hugoniot relation $[\rho v] = 0$ to show that

$$U_{\rm sh} = \frac{[Q]}{[\rho]} \,. \tag{2.1}$$

(b) Taylor expand $[Q]$ in $[\rho]$ to $\mathcal{O}(\epsilon^3)$ and combine with equation (2.1) to show that

$$U_{\rm sh} = Q'_a + \frac{1}{2} Q''_a [\rho] + \mathcal{O}(\epsilon^2) \,. \tag{2.2}$$

(c) Noting that $[Q'_a] = Q''_a[\rho] + \mathcal{O}(\epsilon^2)$, substitute $Q''_a$ into equation (2.2) to show that

$$U_{\rm sh} = Q'_a + \frac{1}{2}[Q'] + \mathcal{O}(\epsilon^2) \,. \tag{2.3}$$

Then write $[Q'] = Q'_b - Q'_a$ to write equation (2.3) as

$$U_{\rm sh} = \frac{1}{2}(Q'_a + Q'_b) + \mathcal{O}(\epsilon^2) \,. \tag{2.4}$$

(d) Note that

$$\begin{aligned} Q' = \frac{d(\rho u)}{d\rho} &= u + \rho \frac{du}{d\rho} \\ &= u + c \\ &= u + c_0 + \frac{B}{2A} u + \mathcal{O}(\epsilon^2) \\ &= c_0 + \beta u \,, \end{aligned} \tag{2.5}$$

where the simple-wave relation $du = \frac{c}{\rho} d\rho$ has been used. Combine equation (2.5) with equation (2.4) to show that

$$U_{\rm sh} = c_0 + \frac{\beta}{2}(u_a + u_b) + \mathcal{O}(\epsilon^2) \,. \tag{2.6}$$

Energy dissipation at a shock front was very involved an is not included in this review. See class notes for the derivation leading to $dE/dt$, which is cubic in the pressure jump. $dT/dt$ is also cubic in the pressure jump. See also the applications to HIFU discussed in class.

# 3 Landau's equal-area rule

(a) Note that the area under a shock is given by

$$A = \int_{u_a}^{u_b} (x - x_{\rm sh}) du \,. \tag{3.1}$$

Write $dA/dt$ using the rule

$$\frac{d}{dt}\int_{u_a}^{u_b} q(u,t)du = q(u_b,t)\frac{du_b}{dt} - q(u_a,t)\frac{du_a}{dt} + \int_{u_a}^{u_b} \frac{\partial q}{\partial t} du.$$

*Hint: let $q$ above $= x - x_{sh}$.*

(b) Noting that $x = x_{\rm sh}$ at $u = u_a$ and $u = u_b$, show that

$$\frac{dA}{dt} = \int_{u_a}^{u_b} \left[\frac{dx}{dt} - \frac{dx_{\rm sh}}{dt}\right] du \,. \tag{3.2}$$

(c) Identify $dx/dt$ in equation (3.2) to be the finite amplitude propagation speed, $c_0 + \beta u + \mathcal{O}(\epsilon^2)$, and identify $dx_{\rm sh}/dt$ to be $U_{\rm sh} = c_0 + \frac{\beta}{2}(u_a + u_b) + \mathcal{O}(\epsilon^2)$, by equation (2.6). Perform the integral in equation (3.2) over $u$ to show that $dA/dt = 0$, i.e., $A =$ constant $= A_+ - A_-$.

# 4 Blackstock's weak-shock method

(a) The retarded shock time is $\tau_{\rm sh} = t_{\rm sh} - x/c_0$. Calculate $d\tau_{\rm sh}/dx$ by define the shock slowness to be $1/U_{\rm sh} = dt_{\rm sh}/dx = [c_0 + \frac{\beta}{2}(u_a + u_b)]^{-1}$. *Answer:*

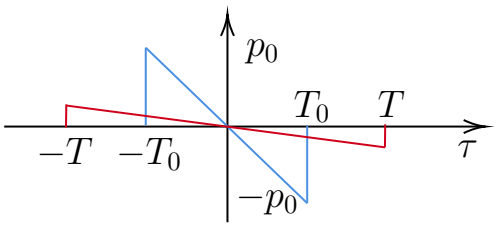
$$\frac{d\tau_{\rm sh}}{dx} = -\frac{\beta}{2\rho_0 c_0^3}(p_a + p_b) + \mathcal{O}(\epsilon^2) \tag{4.1}$$

(b) From where are $p_a$ and $p_b$ obtained?

(c) N-wave example: Use the Blackstock weak shock method to find $p_{\rm sh}(x)$ for the boundary condition

$$f(t) = \begin{cases} -p_0 t, & |t| < T_0 \\ 0, & |t| > T \end{cases} \quad , \tag{4.2}$$

which is prescribed at $x = 0, \phi = \tau$.




# 5 Blackstock's bridging function

(a) From the development of the Fubini solution, which expands the pressure as a Fourier sine series $P(\sigma, \theta) = \sum_{n=1}^{\infty} B_n(\sigma) \sin n\theta$, where $\sigma = x/\bar{x}$ and $\theta = \omega\tau$, it was found that the expansion coefficients $B_n$ are given by the sum $B_n^{(1)} + B_n^{(2)}$, where

$$B_n^{(1)} = -\frac{2}{n\pi} \cos(n\theta) \sin(\Phi) \Big|_{\theta,\Phi=0}^{\theta,\Phi=\pi} = 0 \tag{5.1}$$

$$B_n^{(2)} = \frac{2}{n\pi} \int_{\Phi=0}^{\Phi=\pi} \cos n\theta \cos \Phi d\Phi = \frac{2}{n\sigma} J_n(n\sigma) \tag{5.2}$$

where $\Phi = \theta + \sigma \sin \Phi$. For $\sigma < 1$, why are the limits on $\theta$ and $\Phi$ above equal?

(b) For $\sigma > 1$, what is $\Phi$ when $\theta = \pi$? What are the two possibilities for $\Phi$ at $\theta = 0$? Noting that $P_{\text{sh}} = P_b$, what is the correct choice for $\Phi$?

(c) Given how $\Phi$ and $\theta$ have different limits at $\theta = 0$, how do equations (5.1) and (5.2) change for $\sigma > 1$? (Qualitative answer is sufficient...the math is a bit confusing)

# 6 Nonlinearity in multiple dimensions

(a) 1D spreading is modeled by adding a term $mp/r$ to the LHS of the Burgers equation with no absorption, i.e., $\delta = 0$:

$$\frac{\partial p}{\partial r} + \frac{m}{r} p = \pm \frac{\beta p}{\rho_0 c_0^3} \frac{\partial p}{\partial \tau}. \tag{6.1}$$

where $\tau$ is now $t \mp (r - r_0)/c_0$. What is $m$ for 1D spherical spreading? What is $m$ for 1D cylindrical spreading? What is some restrictions on this formulation?

(b) Introduce

$$q = \left(\frac{r}{r_0}\right)^m p$$

and calculate $\partial p/\partial r$ and $\partial p/\partial \tau$ in terms of $q$.

(c) Write equation (6.1) in terms of $q$. *Answer:*

$$\frac{\partial q}{\partial r} = \pm\left(\frac{r_0}{r}\right)^m \frac{\beta q}{\rho_0 c_0^3}\frac{\partial q}{\partial \tau} \tag{6.2}$$

(d) With the intention of getting rid of the factor of $(r_0/r)^m$ altogether from equation (6.2), choose $z(r)$ such that

$$\frac{\partial q}{\partial r} = \frac{\partial q}{\partial z}\frac{dz}{dr} = \pm\left(\frac{r_0}{r}\right)^m \frac{\partial q}{\partial z}.$$

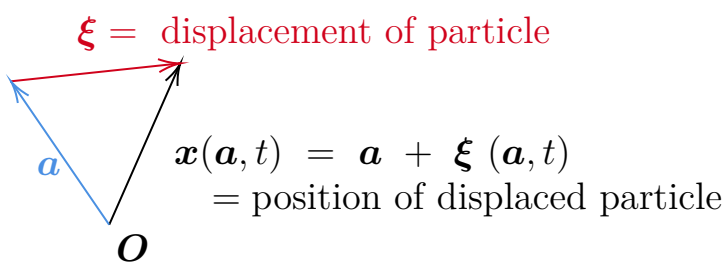
Integrate to find $z$ for $m = 1$ and $m = 1/2$.

(e) Write equation (6.1) in terms of the stretched coordinates $q$ and $z$.

# 7 Radiation force

(a) What is the distinction between Eulerian and Lagrangian coordinates? Why does the distinction dissolve in linear theory?

(b) Let $\boldsymbol{a}$ be the position of a particle at rest, $\boldsymbol{\xi}$ be the displacement of the particle from $\boldsymbol{a}$, and $\boldsymbol{x}$ be the position of the displaced particle:




Then, the transformations between a Lagrangian quantity $q_{\mathrm{L}}(\boldsymbol{a},t)$ (can be a scalar, vector, or tensor) and Eulerian quantity $q_{\mathrm{E}}(\boldsymbol{x},t)$ are

$$q_{\mathrm{L}}(\boldsymbol{a},t) = q_{\mathrm{E}}(\boldsymbol{a},t) + \boldsymbol{\xi}(\boldsymbol{a},t)\cdot\boldsymbol{\nabla} q_{\mathrm{E}}(\boldsymbol{a},t) \tag{7.1}$$

$$q_{\mathrm{E}}(\boldsymbol{a},t) = q_{\mathrm{L}}(\boldsymbol{a},t) - \boldsymbol{\xi}(\boldsymbol{x},t)\cdot\boldsymbol{\nabla} q_{\mathrm{L}}(\boldsymbol{x},t)\,. \tag{7.2}$$

Why can the coordinates in which the gradients in the above equations are evaluated be neglected?

(c) Resolve Westervelt's paradox, which says that for $\dot{X}(t) = u_0 \sin \omega t$ at $x = X(t)$, $u_\mathrm{E} = u_0 \sin(\omega t - kx)$ and $\langle u_\mathrm{E} \rangle = -\langle u^2 \rangle / c_0 = -u_0^2/2c_0$. Do so by calculating $\langle u_\mathrm{L} \rangle$ in Lagrangian coordinates. *Hint: use equation (7.1).*

(d) Calculate the mean excess pressure in Eulerian coordinates. *Hint: Start with the linearized momentum equation*

$$\rho_0 \frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{\nabla} p = -\boldsymbol{\nabla} \mathcal{L} \tag{7.3}$$

*where* $\mathcal{L} = \frac{1}{2}\rho_0 u^2 - p^2/2\rho_0 c_0^2$ *is the Lagrangian (leave everything in terms of* $\mathcal{L}$*). Then let* $\boldsymbol{u} = \boldsymbol{\nabla}\phi$*, i.e., irrotational, and integrate over volume. Call the constant of integration* $g(t)$ *on the right-hand side. Finally take the time average and call* $\langle g(t) \rangle \equiv C$*. Answer:*

$$\langle p_\mathrm{E} \rangle = -\langle \mathcal{L} \rangle + C \tag{7.4}$$

(e) Calculate the mean excess pressure in Lagrangian coordinates by using equation (7.1). *Hint: After using equation (7.1), take the time average, and use the momentum equation to write* $\langle \boldsymbol{\xi} \cdot \boldsymbol{\nabla} p \rangle = -\rho_0 \langle \boldsymbol{\xi} \cdot \partial \boldsymbol{u}/\partial t \rangle$*. Further note that one can write* $\partial^2 \boldsymbol{\xi}/\partial t^2$ *as* $2\partial \boldsymbol{\xi}/\partial t \cdot \partial \boldsymbol{\xi}/\partial t + 2\boldsymbol{\xi} \cdot \partial^2 \boldsymbol{\xi}/\partial t^2$*. Answer:*

$$\langle p_\mathrm{L} \rangle = \langle p_\mathrm{E} \rangle + \rho_0 \langle u^2 \rangle \tag{7.5}$$

(f) Define $V = p^2/2\rho_0 c_0^2$, $K = \rho_0 u^2/2$. Then the energy is $\mathcal{E} = K + V$ and the Lagrangian is $\mathcal{L} = K - V$. By equation (7.5), $\langle p_\mathrm{L} \rangle = \langle p_\mathrm{E} \rangle + 2\langle K \rangle$. Combine this result and the new notation with equation (7.4) to find $\langle p_\mathrm{L} \rangle$.

(g) Show that in the linear limit, the Eulerian and Lagrangian excess pressures are equal.

(h) Show that the Lagrangian radiation pressure $\langle p_\mathrm{L} \rangle$ on a surface normal to and in contact with the fluid motion is constant. What is remarkable about this result? *Hint: Take the time average of Newton's second law, which in Lagrangian coordinates reads* $\rho_0 \partial^2 \xi/\partial t^2 = -\partial p_\mathrm{L}/\partial a$.

(i) Calculate $\langle p_{\mathrm{E}} \rangle$ and $\langle p_{\mathrm{L}} \rangle$ in a standing wave, to accuracy of a constant of integration $C$. The standing wave is given by $p = p_0 \cos kx \sin \omega t$, or equivalently $u = \frac{p_0}{\rho_0 c_0} \sin kx \cos \omega t$. *Hint: Recall from Acoustics I that* $V = p^2/2\rho_0 c_0^2$ *and* $K = \rho_0 u^2/2$ *and write the answer in terms of* $\mathcal{E} = V + K$ *and* $\mathcal{L} = K - V$.

(j) Determine the constant of integration $C$ by invoking the conservation of mass. Specifically, require that

$$\int_x^{x+\lambda} \langle \rho'_{\mathrm{E}} \rangle \, dx = 0, \qquad \text{at } \mathcal{O}(\epsilon^2)\,,$$

where[1] $\rho' = (p - VB/A)/c_0^2$. *Answer:*

$$C = \frac{B}{2A} \frac{p_0^2}{4\rho_0 c_0^2}$$

(k) Given that the radiation force on an object of volume $V$ is $\boldsymbol{F}_{\mathrm{rad}} = -\langle V \boldsymbol{\nabla} p \rangle$, calculate the radiation force exerted on a ping-pong ball of radius $R$ by a standing pressure wave $p(x,t) = p_0 \cos kx \sin \omega t$ in a closed tube. Assume that the ball is perfectly rigid and that $kR \ll 1$. *Hint: Reduce the problem to 1D, i.e.,* $\boldsymbol{F}_{rad} = -\langle V \partial p/\partial x \rangle$ *and assume that the ping-pong ball has sufficient inertia such that Eulerian radiation pressure* $\langle p_{\mathrm{E}} \rangle$ *found in the previous problem can be used. Also note that since the ball is rigid, its volume is constant. All these considerations result in the radiation force being given by*

$$F_{\mathrm{rad}} = -\frac{4\pi R^3}{3} \frac{d\langle p_{\mathrm{E}} \rangle}{dx}$$

(l) Define $\langle \mathcal{P} \rangle$ to be the time-averaged momentum density, given by $\langle \text{momentum/volume} \rangle$. Show that

$$\langle \mathcal{P} \rangle = \frac{\langle \mathcal{E} \rangle}{c_0} \quad \text{for} \quad f(x - c_0 t)$$

and

$$\langle \mathcal{P} \rangle = -\frac{\langle \mathcal{E} \rangle}{c_0} \quad \text{for} \quad f(x + c_0 t).$$

[1] See equation (3-39) of [2].

*Hint: write momentum/volume as $\rho' u$, and use linear relations $\rho' = p/c_0^2$ and $u = p/\rho_0 c_0$. Then note that $\mathcal{E} = p^2/\rho_0 c_0^2$.*

(m) In class, the momentum flux (time-averaged momentum per unit time per unit area) was found to be given by $J = c_0 \langle \mathcal{P} \rangle = \langle \mathcal{E} \rangle$, where $\langle \mathcal{P} \rangle$ is the time-averaged momentum density discussed in the previous part. At a 2-fluid interface, with the first fluid having parameters $\rho_1$ and $c_1$ and the second fluid having parameters $\rho_2$ and $c_2$, the net momentum flux $J$ into the interface was identified to be the time-averaged Lagrangian pressure $\langle p_{\mathrm{L}} \rangle$. Use these relations to find $\langle p_{\mathrm{L}} \rangle$ in terms of the fluid parameters, the incident time averaged energy density $\langle \mathcal{E}_1 \rangle$, and the pressure reflection and transmission coefficients $R$ and $T$. *Hint: Start with $\langle p_{\mathrm{L}} \rangle = c_0 \langle \mathcal{P}_i - \mathcal{P}_r - \mathcal{P}_t \rangle$.*

(n) Given that $F_{\mathrm{rad}} \propto \langle \mathcal{P} \rangle$, is it possible to have acoustic radiation force in the linear limit?

# 8 Streaming

I haven't typed up solutions to the streaming problems.

(a) It was shown that the "full momentum equation" i.e., equation (3-2) of [2], can be written as

$$\frac{\partial(\rho \boldsymbol{u})}{\partial t} - \boldsymbol{F}' + \boldsymbol{\nabla} P = \mu \nabla^2 \boldsymbol{u} + (\mu_B + \mu/3) \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \boldsymbol{u}) \tag{8.1}$$

$$\text{where } -\boldsymbol{F}' = \rho(\boldsymbol{u} \cdot \boldsymbol{\nabla})\boldsymbol{u} + \boldsymbol{u}(\boldsymbol{\nabla} \cdot \rho \boldsymbol{u}) \tag{8.2}$$

Take the time-average of equations (8.1) and (8.2) and denote $\boldsymbol{F} \equiv \langle \boldsymbol{F}' \rangle$ to show that

$$\boldsymbol{F} = \boldsymbol{\nabla} \langle P \rangle - \mu \nabla^2 \langle \boldsymbol{u} \rangle \tag{8.3}$$

$$\boldsymbol{F} = -\langle \rho(\boldsymbol{u} \cdot \boldsymbol{\nabla})\boldsymbol{u} + \boldsymbol{u} \boldsymbol{\nabla} \cdot (\rho \boldsymbol{u}) \rangle \tag{8.4}$$

What assumption has been made about the fluid in equation (8.3)? What is another name for this assumption? Why does the $\langle \partial(\rho \boldsymbol{u})/\partial t \rangle = 0$?

(b) Drop the $\langle\ \rangle$ notation denoting "time average" and let the time averaging be implied on all wave quantities. Letting $P = P_0 + p_1 + p_2$, $\rho = \rho_0 + \rho_1 + \rho_2$, and $\boldsymbol{u} = \boldsymbol{u}_1 + \boldsymbol{u}_2$, the subscripts refer to the order

of the term, show that the $\mathcal{O}(\epsilon^2)$ version of equations (8.3) and (8.4) are

$$\boldsymbol{F}_2 = \boldsymbol{\nabla} p_2 - \mu \nabla^2 \langle \boldsymbol{u}_2 \rangle \tag{8.5}$$
$$\boldsymbol{F}_2 = -\langle \rho_0 (\boldsymbol{u}_1 \cdot \boldsymbol{\nabla}) \boldsymbol{u}_1 + \boldsymbol{u}_1 \boldsymbol{\nabla} \cdot (\rho \boldsymbol{u}_1) \rangle \tag{8.6}$$

(c) Let $p_1 = p_0 e^{-\alpha x} \sin(\omega t - kx)$. Calculate $\boldsymbol{F}_2$ using equation (8.6). *Hint: use the linear relation* $p_1 \simeq \rho_0 c_0 u_1$.

(d) Take the limit of the above result as take the limit as $\alpha \ll k$. *Answer:* $F_{2l} = \alpha p_0^2 / \rho_0 c_0^2$. What does this result say about the nature of acoustic streaming?

(e) In class, it was shown that in the presence of shocks,

$$F_{2f} = \frac{2\beta k P_{\text{sh}}}{3\pi \rho_0^2 c_0^4},$$

the maximum value of which is $2\beta k p_0^3 / 3\pi \rho_0^2 c_0^4$. Show that the ratio of $F_{2f}$ to $F_{2l}$ is $2\Gamma/3\pi$, where $\Gamma = \beta \epsilon k / \alpha$ (the Gol'berg number).

# References


[1] M. F. Hamilton, Lecture notes from Nonlinear Acoustics. University of Texas at Austin, (2023).

[2] M. F. Hamilton and D. T. Blackstock, "Nonlinear Acoustics." Acoustical Society of America, (2008).